Linear Algebra and Regression CheatSheet

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Introduction to Linear Algebra and Regression

Linear Algebra is the mathematical foundation that allows us to work with **vectors**, **matrices**, and **transformations**, which are critical in various fields like machine learning, data science, physics, and engineering. In machine learning, Linear Algebra plays a vital role in the representation and transformation of data, especially in algorithms like **Principal Component Analysis (PCA)** and **neural networks**.

Regression is a statistical technique that models and analyzes relationships between variables. It's one of the most fundamental methods for predictive analytics, where the goal is to predict a dependent variable (output) based on one or more independent variables (inputs). Linear regression, a core type of regression, uses a **linear approach** to model the relationship between variables.

In both fields, mathematical tools like **vectors**, **matrices**, and **eigenvalues** are essential for performing operations like optimizations (e.g., **Gradient Descent**) and reducing the dimensionality of large datasets.

In summary:
Linear Algebra provides tools to work with multi-dimensional data.
Regression helps us model and predict relationships between variables.
Both together are indispensable in modern data-driven industries, especially for machine learning and statistical analysis.

Vectors

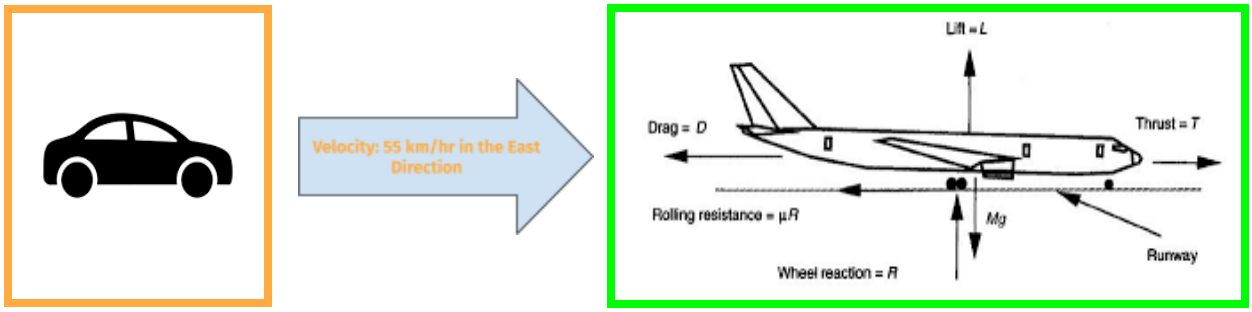
What are Vectors?

A vector is a mathematical object that represents both **magnitude** and **direction**. Unlike a scalar, which only has magnitude (e.g., temperature, mass), a vector tells us both how large something is and in which direction it points. Vectors are often used to describe physical quantities like **force**, **velocity**, and **displacement**, making them essential in fields like physics, engineering, and data science.

Key Features of Vectors:

Magnitude: The size or length of the vector.
Direction: The orientation of the vector in space.

Examples of Vectors



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Decomposition of Vectors

Decomposition of a vector involves **breaking it down** into its components along the axes of a coordinate system (usually **x**, **y**, and **z** in **3D** space). This allows us to express a vector as the sum of its individual parts, which makes it easier to perform operations like addition, subtraction, and projections.

How it Works:

Any vector v in 2D or 3D space can be decomposed into components along the x, y, and z axes:

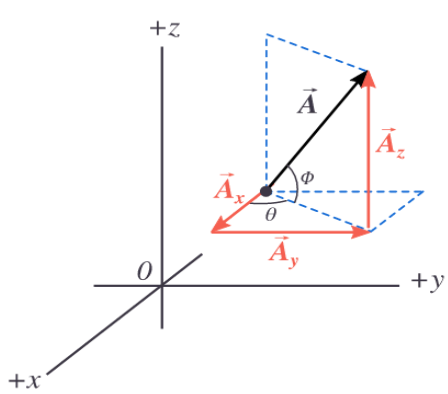
$$\mathbf{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$$

Where:

- v_x, v_y, and v_z are the scalar components of the vector along the x, y, and z axes.
- $\hat{i}, \hat{j}, \hat{k}$ are the unit vectors along the x, y, and z axes.

Example:

A ball traveling in a 3-D space. It's velocity can be decomposed into x, y, and z components and represented as follows:



$$\boxed{\vec{A} = 2\hat{i} + 3\hat{j} + 5\hat{k}}$$

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Vector Operations

Addition and Subtraction

Vectors can be added or subtracted component-wise.

$$\mathbf{a} = \begin{pmatrix} a_x \\ a_y \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} b_x \\ b_y \end{pmatrix}$$

Addition=> $$\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_x + b_x \\ a_y + b_y \end{pmatrix}$$

Subtraction=> $$\mathbf{a} - \mathbf{b} = \begin{pmatrix} a_x - b_x \\ a_y - b_y \end{pmatrix}$$

Two vectors acting in **opposite directions** Having different magnitudes Net force → west direction 3 N (in the direction of higher force) V1 = 7N V2 = -10N V1 + V2 = -3N	Two vectors acting in **same direction** Having different magnitudes Net force → east direction 17 N (same direction of the forces) V1 = 7N V2 = 10N V1 + V2 = 17N

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Scalar Multiplication

Scalar multiplication involves multiplying a **vector** by a **scalar** (a real number). This operation changes the **magnitude** of the vector but keeps its **direction the same** (**unless the scalar is negative**, which reverses the direction).

If $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}$ is a vector and k is a scalar, then the scalar multiplication of k and $\mathbf{v}$ is calculated by multiplying each component of the vector by k:

$$k\mathbf{v} = \begin{pmatrix} kv_x \\ kv_y \\ kv_z \end{pmatrix}$$

If $k > 1$, the vector's magnitude increases.

If $0 < k < 1$, the vector's magnitude decreases.

If $k = 0$, the vector becomes the zero vector $\mathbf{0}$.

If $k < 0$, the vector's direction reverses.

Norm (Magnitude)

The norm (or magnitude) of a vector is its length and is calculated using the **Euclidean distance formula**:

$$\boxed{\|a\| = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}}$$

This gives the length of the vector.

Distance

The distance between two vectors a and b is the length of the vector $a-b$, calculated as:

$$\boxed{d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \cdots + (u_n - v_n)^2}}$$

Dot Product

The dot product measures how much two vectors are aligned. For vectors a and b:

$$\boxed{\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y}$$

Angle Between Vectors

The angle θ between two vectors can be calculated using the dot product formula:

$$\boxed{\cos(\theta) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}}$$

Projection

The projection of vector a onto b is the component of a that lies along b. It is given by:

$$\boxed{\text{Proj}_b a = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2}\mathbf{b}}$$

Matrix Operations 1

A matrix is a rectangular array of numbers arranged in **rows** and **columns**. It is a powerful mathematical tool used to represent and solve systems of linear equations, transformations in space, and operations in computer science, physics, and engineering.

Matrix Notation:

A matrix is usually denoted by a capital letter (e.g., A), and its elements are organized in rows and columns inside brackets. The general form of an $m \times n$ matrix is:

$$\boxed{A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}}$$

Where:

- m is the number of rows.
- n is the number of columns.

Types of Matrices:

Square Matrix: A matrix where the number of rows equals the number of columns **(**m **=** n**)**.

Row Matrix: A matrix with only one row **(1 ×** n**)**.

Column Matrix: A matrix with only one column **(**m **× 1)**.

Zero Matrix: A matrix where all elements are **zero**.

Diagonal Matrix: A square matrix where **all non-diagonal elements are zero**.

Identity Matrix: A diagonal matrix where **all diagonal elements are 1**.

Element Notation:

The element in the **i-th** row and **j-th** column of a matrix **A** is denoted as $\mathbf{a_{ij}}$.

Matrices are fundamental in representing linear transformations, solving systems of equations, and performing operations in multiple dimensions.

Matrix Addition

Two matrices can be added if they have the **same dimensions** (i.e., **the same number of rows and columns**). The addition is performed by adding corresponding elements from both matrices.

If A and B are two **m×n** matrices, the sum **C=A+B** is given by:

$$C = A + B = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{pmatrix}$$

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

$$A + B = \begin{pmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}$$

Scalar Multiplication

In scalar multiplication, each element of a matrix is multiplied by a scalar (a single number). If **A** is an **m×n** matrix and **k** is a scalar, then the result of multiplying matrix **A** by scalar **K** is:

$$kA = \begin{pmatrix} k \cdot a_{11} & k \cdot a_{12} & \cdots & k \cdot a_{1n} \\ k \cdot a_{21} & k \cdot a_{22} & \cdots & k \cdot a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ k \cdot a_{m1} & k \cdot a_{m2} & \cdots & k \cdot a_{mn} \end{pmatrix}$$

Example:

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $k = 3$, then:

$$kA = 3 \times \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix}$$

Matrix Multiplication

Matrix multiplication is more complex than addition or scalar multiplication. Two matrices can be multiplied if the **number of columns** of the first matrix equals the **number of rows** of the second matrix.

If **A** is an **m×p** matrix and **B** is a **p×n** matrix, then the product **C=AB** is an **m×n** matrix. The element $\mathbf{C_{ij}}$ in matrix **C** is calculated as the dot product of the i-th row of matrix **A** and the j-th column of matrix **B**:

$$C = AB = \begin{pmatrix} \sum a_{1k}b_{k1} & \sum a_{1k}b_{k2} & \cdots & \sum a_{1k}b_{kn} \\ \sum a_{2k}b_{k1} & \sum a_{2k}b_{k2} & \cdots & \sum a_{2k}b_{kn} \\ \vdots & \vdots & \ddots & \vdots \\ \sum a_{mk}b_{k1} & \sum a_{mk}b_{k2} & \cdots & \sum a_{mk}b_{kn} \end{pmatrix}$$

Example:

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$, then:

$$AB = \begin{pmatrix} (1 \cdot 5 + 2 \cdot 7) & (1 \cdot 6 + 2 \cdot 8) \\ (3 \cdot 5 + 4 \cdot 7) & (3 \cdot 6 + 4 \cdot 8) \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$

Determinant Calculation

The **determinant** of a matrix is a scalar value that provides important information about the matrix, such as whether it is invertible and the scaling factor of the transformation it represents. Determinants are widely used in linear algebra, especially for solving systems of equations, finding inverses, and determining eigenvalues.

Determinant of a 1x1 Matrix:

For a 1×1 matrix $A = \begin{pmatrix} a \end{pmatrix}$, the determinant is simply the value of the single element:

$$\det(A) = a$$

Example:

If $A = \begin{pmatrix} 5 \end{pmatrix}$, then:

$$\det(A) = 5$$

Determinant of a 2x2 Matrix:

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the determinant is calculated as:

$$\det(A) = ad - bc$$

Example:

Let $A = \begin{pmatrix} 3 & 8 \\ 4 & 6 \end{pmatrix}$, then:

$$\det(A) = (3 \times 6) - (8 \times 4) = 18 - 32 = -14$$

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Determinant of a 3x3 Matrix:

For a 3×3 matrix $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, the determinant is calculated using cofactor expansion along the first row:

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Example:

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{pmatrix}$, then:

$$\det(A) = 1 \times (4 \times 6 - 5 \times 0) - 2 \times (0 \times 6 - 5 \times 1) + 3 \times (0 \times 0 - 4 \times 1)$$

$$\det(A) = 1 \times 24 - 2 \times (-5) + 3 \times (-4) = 24 + 10 - 12 = 22$$

Rank of a Matrix

The **rank** of a matrix is a fundamental concept in linear algebra that represents the maximum number of **linearly independent rows or columns** in the matrix. It provides insight into the matrix's properties, such as whether a system of equations has a unique solution, no solution, or infinitely many solutions.

Definition:

The rank of a matrix is the dimension of the vector space spanned by its rows (row rank) or columns (column rank). For any matrix, the row rank and column rank are always equal.

- **Full Rank**: A matrix has full rank if its rank is equal to the smaller of its number of rows or columns.

- **Reduced Rank**: If a matrix has fewer linearly independent rows or columns than its total number of rows or columns, its rank is reduced.

How to Determine the Rank:

1. **Row Echelon Form**: One way to find the rank is to reduce the matrix to its row echelon form (or reduced row echelon form) using Gaussian elimination. The number of non-zero rows in this form gives the rank of the matrix.

2. **Independent Rows/Columns**: Count the number of linearly independent rows or columns. A row or column is linearly independent if it cannot be written as a linear combination of other rows or columns.

Examples:

- **Full-Rank Matrix**: A 3×3 matrix is said to have full rank if its rank is 3, meaning all three rows (or columns) are linearly independent.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

  This matrix has a rank of 2 because the third row is a linear combination of the first two rows, meaning it is not linearly independent.

- **Rank-Deficient Matrix**: A 3×3 matrix where one or more rows (or columns) can be expressed as a linear combination of the others has a rank less than 3.

Summary:

- The **rank** of a matrix is the maximum number of linearly independent rows or columns.
- It indicates the dimension of the vector space spanned by the matrix's rows or columns.
- A matrix has **full rank** if all rows (or columns) are independent, and **reduced rank** if they are not.

Reduced Row Echelon Form (RREF)

Pivot:

- A matrix element $\mathbf{a_{ij}}$ is a pivot if all entries to the left and below it are zero.

Row Echelon Form (REF):

- A matrix where all non-zero rows have a pivot.
- Any pivot is to the right of previous rows' pivots.
- All-zero rows are at the bottom.

Reduced Row Echelon Form (RREF):

- A matrix in row echelon form with additional conditions:
  - Pivots are all 1.
  - Columns containing pivots have zeros in other rows.

Example of RREF:

Given the matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{pmatrix}$$

After applying row operations to convert it to **RREF**:

$$\text{RREF}(A) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

- The leading entries (pivots) are 1s.
- Each pivot has zeros in its column.
- The bottom row is all zeros.

This matrix is now in **Reduced Row Echelon Form (RREF)**.

Singular Matrix

A **singular matrix** is a square matrix that **does not have an inverse**. This happens when the **determinant** of the matrix is zero **(det(A)=0)**. Singular matrices are important because they indicate that the system of linear equations represented by the matrix has no unique solution (i.e., it could have either no solutions or infinitely many solutions).

Key Characteristics of a Singular Matrix:

1. **Determinant is Zero**: For any square matrix A, **if det(A)=0, then A is singular.**
2. **Not Invertible**: A singular matrix does not have an inverse.
3. **Dependent Rows/Columns**: In a singular matrix, at least one row or column is a linear combination of the others, meaning the rows or columns are linearly dependent.
4. **Rank is Less than Full**: The rank of a singular matrix is less than its size.

Example:

Consider the matrix:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

The determinant of A is:

$$\det(A) = (1 \times 4) - (2 \times 2) = 4 - 4 = 0$$

Since $\det(A) = 0$, the matrix is **singular**.

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Matrix Operations 2

Transpose of a Matrix

The **transpose** of a matrix is obtained by **interchanging its rows and columns**.

Example:

Let A be a 2×3 matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

The transpose of A is:

$$A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

Now A^T is a 3×2 matrix.

Properties of Transpose:

1. $(A^T)^T = A$: The transpose of the transpose is the original matrix.
2. $(A + B)^T = A^T + B^T$: The transpose of a sum of two matrices is the sum of their transposes.
3. $(kA)^T = kA^T$: The transpose of a scalar multiple of a matrix is the scalar multiple of the transpose.
4. $(AB)^T = B^T A^T$: The transpose of the product of two matrices is the product of their transposes in reverse order.

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Inverse of a Matrix

The **inverse** of a matrix A is another matrix, denoted as A^{-1}, such that when multiplied with A, it results in the **identity matrix** I. The identity matrix is the matrix equivalent of the number 1 in multiplication for real numbers.

For a square matrix A, its inverse A^{-1} satisfies:

$$A \cdot A^{-1} = A^{-1} \cdot A = I$$

Where I is the identity matrix of the same size as A.

Conditions for a Matrix to Have an Inverse:

- **Square Matrix**: Only square matrices (matrices with the same number of rows and columns) can have inverses.

- **Non-Singular Matrix**: A matrix must be non-singular, meaning its **determinant** is non-zero (det(A)≠0).

How to Calculate the Inverse:

How to Calculate the Inverse:

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the inverse is given by:

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Where $\det(A) = ad - bc$ is the determinant of the matrix.

Properties of Inverse

Properties of Inverses:

1. **Uniqueness**: The inverse of a matrix, if it exists, is unique.
2. **Inverse of a Product**: $(AB)^{-1} = B^{-1}A^{-1}$.
3. **Inverse of a Transpose**: $(A^T)^{-1} = (A^{-1})^T$.
4. **Inverse of an Identity Matrix**: The identity matrix is its own inverse, $I^{-1} = I$.

When a Matrix Does Not Have an Inverse:

A matrix is **singular** and does not have an inverse if:

- Its determinant is zero ($\det(A) = 0$).
- The rows or columns are linearly dependent (i.e., one row/column is a scalar multiple of another).

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Eigenvalues and Eigenvectors

Eigenvalues and **eigenvectors** are fundamental concepts in linear algebra used to understand linear transformations. They are widely applied in fields like machine learning, physics, and engineering, especially for tasks like dimensionality reduction (e.g., **Principal Component Analysis**).

Definition:

For a square matrix **A**, an **eigenvector** is a non-zero vector **V** such that when **A** acts on **V**, the result is a scalar multiple of **V**. The scalar is called the **eigenvalue** associated with the eigenvector.

$$A\mathbf{v} = \lambda\mathbf{v}$$

Where:

- A is the matrix.
- $\mathbf{v}$ is the eigenvector.
- λ is the eigenvalue.

Eigenvalue Equation:

To find eigenvalues, solve the equation:

$$\det(A - \lambda I) = 0$$

Where I is the identity matrix, and λ represents the eigenvalues of the matrix.

How to Calculate Eigenvalues and Eigenvectors:

1. **Eigenvalues**: Solve the characteristic equation $\det(A - \lambda I) = 0$ for λ.
2. **Eigenvectors**: For each eigenvalue λ, solve $(A - \lambda I)\mathbf{v} = 0$ to find the corresponding eigenvector $\mathbf{v}$.

Example:

Given matrix $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$:

1. **Find Eigenvalues**: Solve $\det(A - \lambda I) = 0$:

$$\det \begin{pmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{pmatrix} = 0$$

   Expanding the determinant:

$$(4 - \lambda)(3 - \lambda) - (1 \times 2) = 0$$

   Simplifying:

$$\lambda^2 - 7\lambda + 10 = 0$$

   Solving the quadratic equation gives the eigenvalues: $\lambda_1 = 5$, $\lambda_2 = 2$.

2. **Find Eigenvectors**: For $\lambda_1 = 5$, solve $(A - 5I)\mathbf{v} = 0$:

$$\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$$

   Solving this gives the eigenvector corresponding to $\lambda_1 = 5$ as $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

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Geometric Interpretation of Eigenvalues and Eigenvectors

The **geometric interpretation** of eigenvalues and eigenvectors revolves around how a matrix transformation affects the direction and magnitude of vectors. In simple terms, eigenvectors represent directions in space that remain unchanged (except for scaling) when a linear transformation (represented by a matrix) is applied. The eigenvalues represent the **scaling factors** by which the eigenvectors are stretched or compressed.

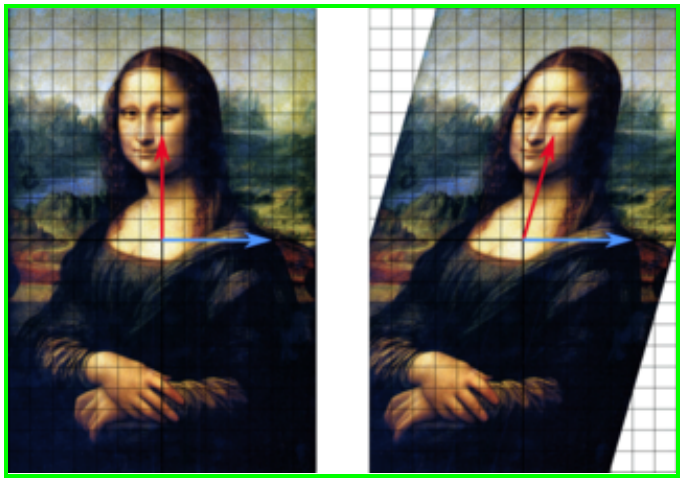
Key Concepts:

- **Eigenvectors**: These are the directions that do not change direction when the matrix transformation is applied. Instead, they only get scaled by the eigenvalue.
- **Eigenvalues**: These are the factors by which the eigenvectors are stretched or compressed during the transformation.

Visualizing in 2D:

Consider a 2D matrix transformation that rotates, stretches, or compresses vectors in a plane:

- A vector that is **not** an eigenvector may change both its direction and magnitude when the matrix transformation is applied.
- A vector that **is** an eigenvector will not change its direction; it will only be scaled (stretched or compressed) by the corresponding eigenvalue.



Application in Transformations:

- **Stretching/Scaling**: Eigenvalues greater than 1 cause the eigenvector to stretch in the direction of the vector.
- **Compression**: Eigenvalues between 0 and 1 compress the eigenvector.
- **Reflection**: Negative eigenvalues reflect the eigenvector across the origin (reverse direction).
- **No Change**: An eigenvalue of 1 leaves the vector unchanged in both magnitude and direction.

Properties:

1. Eigenvalues of real symmetric matrices are always real.
2. The eigenvalues of skew-symmetric matrices is undefined
3. Non-singular matrices have all non-zero eigenvalues
4. The sum of the eigenvalues of a matrix is equal to the trace of the matrix

5. Eigenvectors with Distinct Eigenvalues are Linearly Independent
6. If **A** is a singular matrix, then $\lambda = 0$ is an eigenvalue of A

Applications:

- **Principal Component Analysis (PCA)**: Eigenvalues and eigenvectors are used in PCA to reduce the dimensionality of datasets by identifying the most important directions (principal components).
- **Stability Analysis**: In systems of differential equations, eigenvalues are used to study stability and behavior over time.
- **Quantum Mechanics**: Eigenvalues correspond to measurable quantities like energy levels in quantum systems.
- Eigenvalues are used in regularization to prevent overfitting.
- Eigenvectors and eigenvalues are used in facial recognition techniques such as EigenFaces.

Linear Regression

Why Linear Regression?

- **Foundation for Machine Learning:** Linear regression is fundamental in understanding more complex machine learning models. It provides the basis for comprehending how algorithms make predictions, which is essential before moving on to more intricate models like neural networks or ensemble methods.
- **Interpretability:** One of the key strengths of linear regression is its interpretability. Each coefficient in the model provides clear insights into how much each feature contributes to the outcome. This makes it invaluable for scenarios where understanding the influence of variables is crucial.
- **Benchmarking Tool:** Linear regression often serves as a baseline model in predictive modeling. Due to its simplicity and speed, it is used to establish a performance benchmark for more complex algorithms, making it easier to assess the incremental value added by advanced models.
- **Versatility Across Domains:** Its principles are applicable across various fields such as economics, biology, engineering, and more. This versatility makes it a valuable tool for professionals across different sectors, enhancing their skills in applying statistical methods to real-world problems.

Linear Regression Examples

Linear regression is a versatile tool used across various industries and scenarios to predict outcomes based on linear relationships. Here are some practical examples illustrating its applications:

1. Real Estate Pricing:

- **Objective**: Predicting the selling prices of homes based on features such as square footage, number of bedrooms, location, and other amenities.
- **Data Used**: Historical data of sold properties, including their features and sold prices.
- **Model**: A linear regression model where the response variable is the home price and the predictors include the home's size, age, location, and number of rooms.

2. Credit Scoring:

- **Objective**: Estimating the creditworthiness of individuals based on their financial history, income level, debt-to-income ratio, and other financial metrics.
- **Data Used**: Financial records, credit history, employment status, and income details.
- **Model**: Linear regression can be used to assign a credit score that predicts the likelihood of defaulting on a loan based on financial predictors.

Statistical Modeling Simplified

Statistical modeling helps us understand and predict events by analyzing data. Here's a straightforward breakdown:

What is Statistical Modeling?

- **Simple Idea**: It's like creating a recipe that predicts results based on ingredients (data).
- **Use**: It helps in making predictions, like forecasting sales or understanding trends.

Basic Elements:

- **Variables**:
    1. **Dependent Variables**: What you want to predict (like sales).
    2. **Independent Variables**: Factors that influence the prediction (like price or marketing spend).

- **Steps to Build a Model**:
    1. **Choose the Model**: Pick a math formula that seems right for the data.
    2. **Estimate Parameters**: Find the best numbers to fit the formula to your data.
    3. **Check the Fit**: See if the model's predictions make sense.
    4. **Validate**: Make sure the model works well with new, unseen data.
    5. **Interpret**: Understand what the model's results mean for your question.

Types of Models:

- **Linear Regression**: Predicts a number (like future sales).
- **Logistic Regression**: Predicts yes/no outcomes (like if a customer will buy).
- **Time Series Analysis**: Predicts trends over time (like stock prices).

Why Use It?

- **Make Decisions**: Helps businesses decide on the next steps.
- **Understand Patterns**: Shows what's important and what's not.

Challenges:

- **Data Quality**: Bad data can lead to wrong predictions.
- **Complexity**: Too complex models can confuse more than clarify.
- **Assumptions**: Every model assumes something; if those assumptions are wrong, the model might be too.

Use Cases:

- **Business**: Planning inventory, setting prices.
- **Healthcare**: Predicting disease outbreaks.
- **Environment**: Forecasting weather patterns.

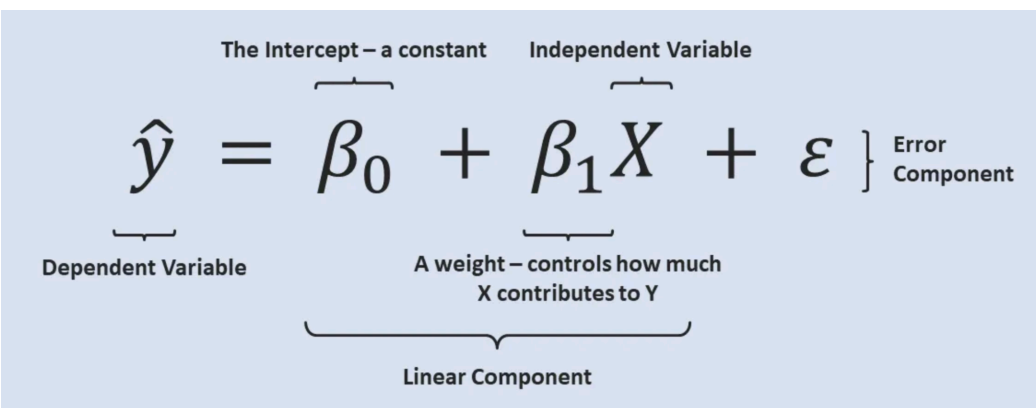
Simple Linear Regression

Simple Linear Regression is a statistical method used to understand the relationship between two continuous variables. It's a straightforward approach for predicting the value of one variable based on the value of another.

Basic Concept:

- **What It Does**: Predicts a dependent variable (like sales) based on a single independent variable (like advertising spend).

Formula:



- Y: Dependent variable (what you want to predict)
- X: Independent variable (what you use to make the prediction)
- β_0: Y-intercept (the starting value when X=0)
- β_1: Slope of the line (how much Y changes with a one-unit change in X)
- ϵ: Error term (the difference between predicted and actual values)

Finding the Best Regression Parameters

The "**best set of values**" for the regression parameters, whether they are denoted as **a** and **b**, $\mathbf{w_0}$ and $\mathbf{w_1}$, or $\boldsymbol{\beta_0}$ and $\boldsymbol{\beta_1}$, are those that minimize the error between the predicted values and the actual data points. This process of minimizing the error helps to position the regression line "at the center of your data," effectively balancing the data around the line.

Concept of 'Center of Your Data'

The idea of positioning the regression line at the **"center of your data"** refers to the concept of minimizing the discrepancies or distances between the observed data points and the points predicted by the model along the **Y-axis**. This concept is operationalized through the method of least squares.

Ordinary Least Squares Regression (OLS Regression)

Ordinary Least Squares **(OLS)** Regression is a method used in linear regression that estimates the parameters of the model by minimizing the sum of the squared errors (or residuals). This method is one of the most basic yet powerful tools in statistical modeling for predicting a continuous variable.

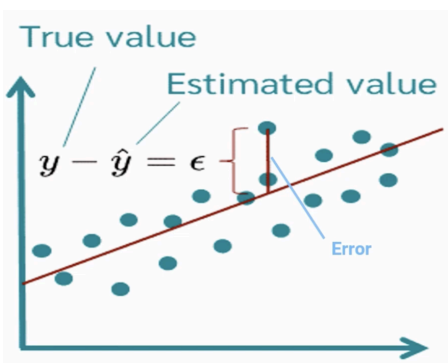
Objective of OLS Regression

The objective of OLS regression is to find the values of the regression coefficients that minimize the sum of the squared differences between the observed dependent variable values and those predicted by the model. This is known as the least squares criterion.

Mathematical Formulation

Given a dataset with **n** observations **(x_i,y_i)**, where **x_i** are the independent variable values and **y_i** are the observed dependent variable values, the goal is to fit a line: **$Y = \beta_0 + \beta_1 X + \epsilon$**

- Y is the dependent variable.
- X is the independent variable.
- β_0 is the intercept of the line.
- β_1 is the slope of the line.
- ϵ represents the error term for each observation.



$$\sum_{i=1}^{n}(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)^2$$

B_0 and **B_1** that minimize the function above are given by:

$$\beta_0 = \frac{\sum y \sum x^2 - \sum x \sum xy}{\sum x^2 - (\sum x)^2} \qquad \beta_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

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Parameter estimates

Some books use linear algebra language, but the calculus remains the same:

Matrix Formulation: For computational efficiency, especially with large datasets, OLS can be formulated as a matrix problem and solved using matrix operations:

$$\beta = (X^T X)^{-1} X^T Y$$

Where X is the matrix of input features, including a column of ones for the intercept, and Y is the vector of output values.

Parameter Estimates Using Gradient Descent

Gradient Descent is a widely used optimization algorithm in machine learning and statistics for finding the minimum of a function, specifically useful for training models like linear regression. It's particularly beneficial when the dataset is too large to fit into memory or when using more complex models that cannot be solved analytically.

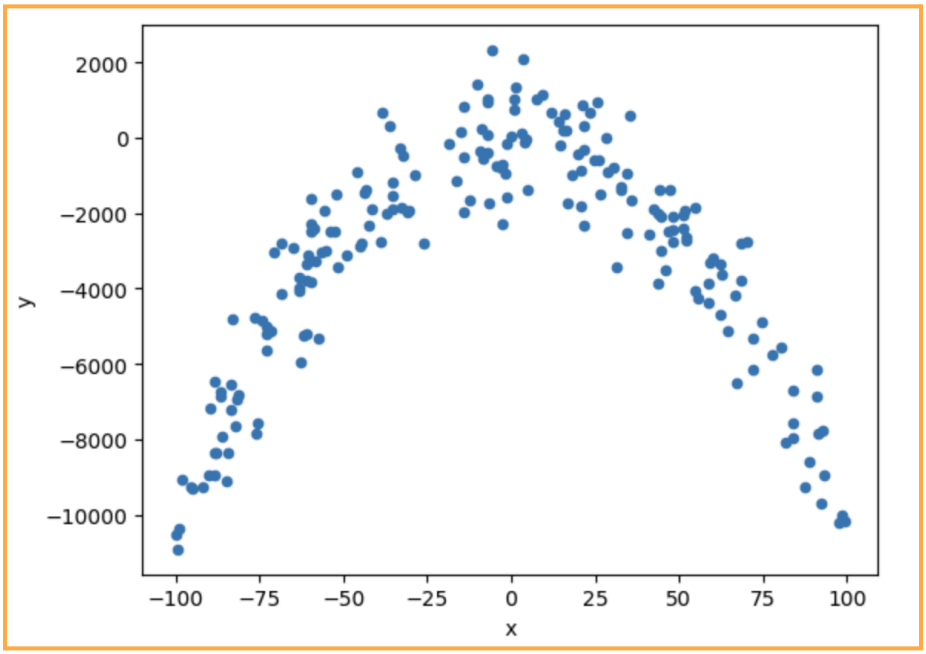
Concept of Gradient Descent:

Gradient Descent helps find the optimal parameters (**β_0**, **β_1** in linear regression) by iteratively moving towards the minimum of the cost function, which in the case of linear regression is typically the sum of squared residuals.

Non-linear Relationships

When analyzing data, it's common to start with a linear model because of its simplicity and interpretability. However, not all relationships between variables are linear. When a pattern in the data does not align with a straight line, a linear model (straight line formula) may not adequately capture the relationship between the variables. This requires exploring non-linear models that can better fit the data.

When there is a pattern, but is not linear. The Straight line formula may not be adequate:



In summary, when a straight line is not adequate to describe the relationship between variables, exploring non-linear models or transformations is essential. This approach can lead to better modeling and understanding of complex patterns in the data.

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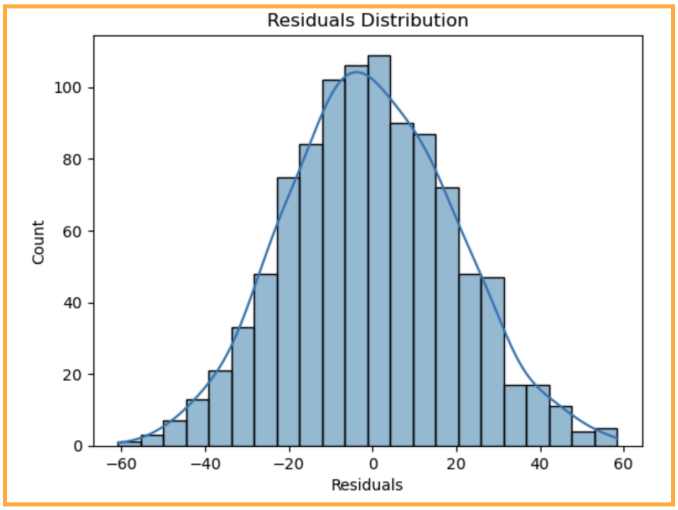
Normality of Errors

In regression analysis, the assumption of normality of errors pertains to the distribution of the residuals, which are the differences between observed values and the values predicted by the model. Ensuring that residuals follow a normal distribution is crucial for the reliability of many statistical tests related to regression coefficients.

Why is Normality Important?

Statistical Inference: Many tests, including **t-tests** for coefficient significance and **F-tests** for model fit, assume that the residuals are normally distributed. This assumption allows for the derivation of confidence intervals and hypothesis tests.
Validity of Results: Normality ensures that the estimates obtained are unbiased and that the associated statistical tests are valid.

When you look at your errors (aka Residuals), they should be concentrated around 0 and follow a normal distribution. In other words, big errors should be rare and follow a similar frequency in both sides (positive and negative errors).



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Uncorrelated Errors

If there is any systematic pattern happening at the errors, it means your formula is missing it. The error of observation n should not predict the next error on (or any other error)

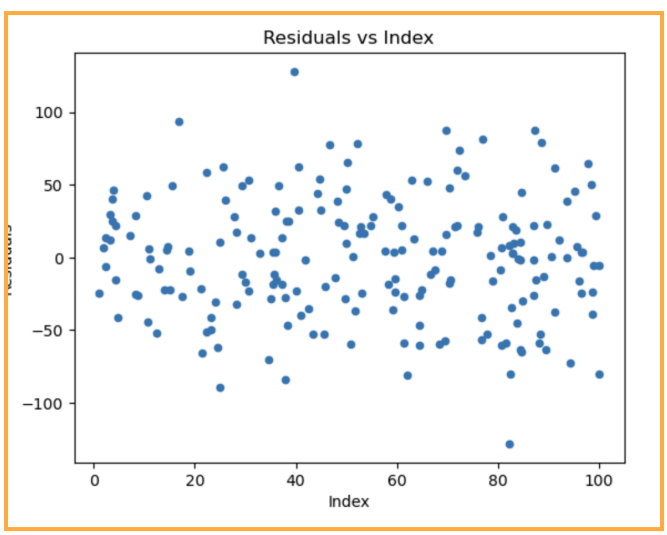
Importance of Uncorrelated Errors

Reliability of Estimates: When errors are correlated, the standard errors of the regression coefficients can be underestimated or overestimated, leading to unreliable hypothesis tests and confidence intervals.

Model Integrity: Correlated errors often indicate model misspecification, such as omitting a relevant variable or an incorrect functional form, which can bias the estimates of the model parameters.

Implications of Ignoring Correlated Errors

- **Poor Predictive Performance**: A model that incorrectly assumes independent errors may perform poorly on new, unseen data, particularly if the data generation process involves dependencies not captured in the model.



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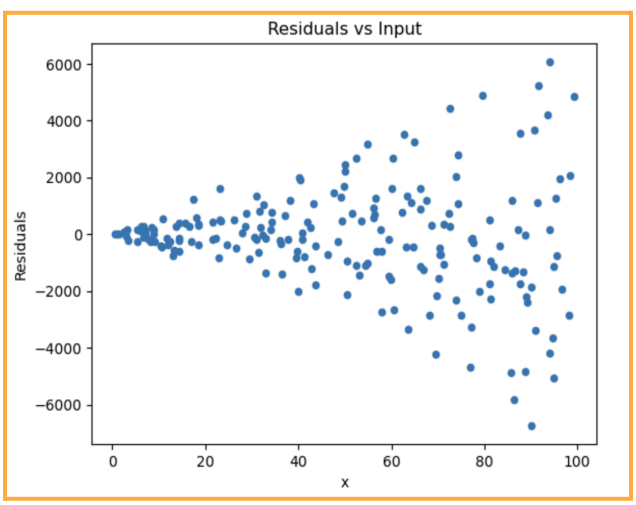
Homoscedasticity

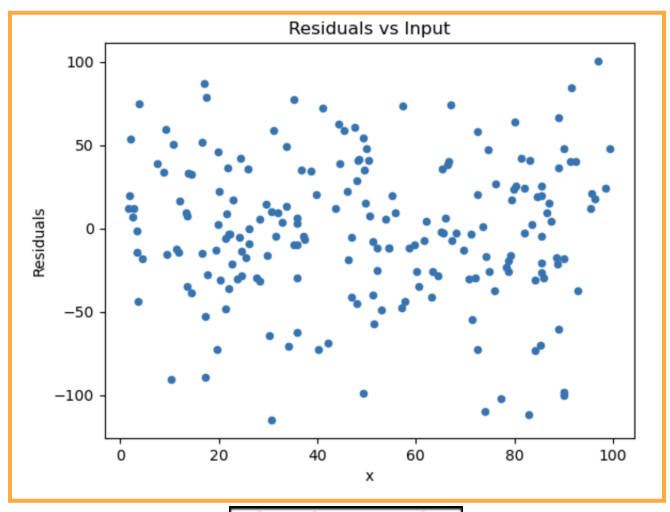
Homoscedasticity is a fundamental assumption in regression analysis that pertains to the consistency of variance across the range of predicted values. It ensures that the spread of the residuals (errors) is constant across all levels of the independent variables.

Understanding Homoscedasticity

- **Definition**: The term **"homoscedasticity,"** often intimidating at first, simply means that the variance of the residuals is the same across all values of the independent variables. In other words, the size or magnitude of the prediction errors does not depend on the value of the predictors.
- **Contrast with Heteroscedasticity**: The opposite condition, **heteroscedasticity**, occurs when the variance of the residuals varies along with the independent variables. This often results in a fan or cone-shaped pattern in the residual plot, where the spread of residuals increases or decreases as the value of the predictor increases.



Looks like a vuvuzela



Looks good.

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Measuring Goodness-of-Fit

Goodness-of-fit measures tell you how well your model's predictions match the actual data.

R^2 and Sum of Squares Total (SST)

Sum of Squares Total (SST) is a key component in calculating R^2, the Coefficient of Determination, which measures the proportion of variance in the dependent variable that is predictable from the independent variables in a regression model. SST is critical for understanding how well a model performs in terms of explaining the variations in the data.

Definition of SST

- **Sum of Squares Total (SST)** represents the total variance in the dependent variable, and it is used as the denominator in the calculation of R^2. It quantifies the total deviation of each observed value from the mean of the dependent variable.

Calculation of SST

SST is calculated as follows:

$$SST = \sum_{i=1}^{n} (y_i - \overline{y})^2$$

Where:

- y_i = Each observed value of the dependent variable
- $\overline{y}$ = Mean of the dependent variable
- n = Number of observations

R^2 and Sum of Squares Errors (SSE)

The **Sum of Squares Errors (SSE)**, also known as the **Sum of Squares of Residuals (SSR)**, is a critical measure in regression analysis used to quantify the discrepancy between the data and an estimation model. SSE is a key component in the calculation of R^2, the Coefficient of Determination, which assesses the proportion of variance in the dependent variable explained by the independent variables.

Definition of SSE

- **Sum of Squares Errors (SSE)** measures the total squared deviation of the observed values from their predicted values, indicating the error produced by the model.

Calculation of SSE

SSE is calculated as follows:

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Where:

- y_i = Each observed value of the dependent variable.
- $\hat{y}_i$ = Each predicted value from the regression model.
- n = Number of observations.

R^2 Formula

Because SST is the numerator of var(y), we can interpret R2 as the amount of variance of y that your model captures.

$$\boxed{R^2 = 1 - \frac{SSE}{SST}}$$

Multiple Linear Regression

Multiple Linear Regression is an extension of simple linear regression, enabling you to predict an outcome based on multiple predictors. This approach is incredibly useful when a single variable does not adequately explain the dependent variable, and multiple factors are at play. Let's explore the case where we have two input variables.

Basic Concept

In multiple linear regression with two inputs, the model considers how both independent variables contribute to the dependent variable. The equation for this relationship is:

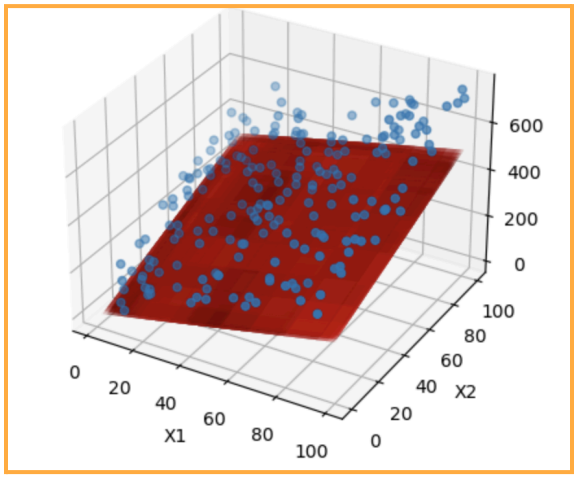
$$Y=\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

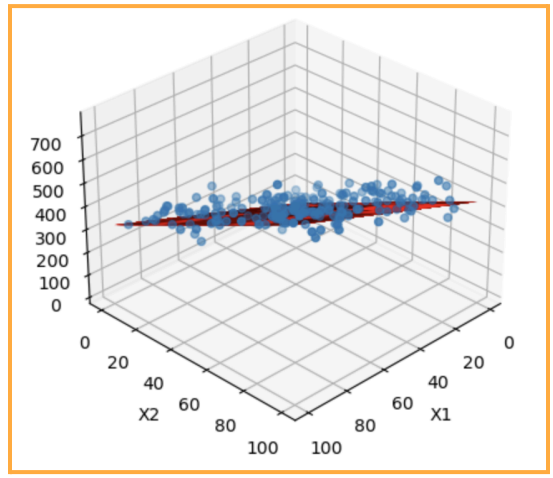
Where:

- **Y** is the dependent variable you want to predict.
- X_1 and X_2 are the independent variables (inputs).
- β_0 is the intercept of the regression plane with the **Y**-axis.

- β_1 and β_2 are the coefficients that represent the impact of each independent variable on the dependent variable.
- ϵ is the error term, accounting for the variability in **Y** not explained by X_1 and X_2.

For this case the simplest formula is the straight plane:





Parameter estimates

When conducting multiple linear regression, determining the best coefficients (**β** values) for the model involves selecting a suitable estimation method. The two most commonly used methods are Ordinary Least Squares (OLS) and Gradient Descent. Each method has its strengths and is suited to different situations depending on the data and computational resources.

$$\hat{\beta} = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{y}$$

$\hat{\beta}$ = ordinary least squares estimator

X = matrix regressor variable X

$\top$ = matrix transpose

y = vector of the value of the response variable

Multiple Linear Regression with Three Inputs

Multiple Linear Regression (MLR) is a statistical technique that allows you to model the relationship between a dependent variable and multiple independent variables. When you have three input variables, MLR helps to understand how each input, or a combination of inputs, influences the outcome.

Formulating the Model

When extending the model to include three independent variables, the regression equation becomes:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$$

Where:

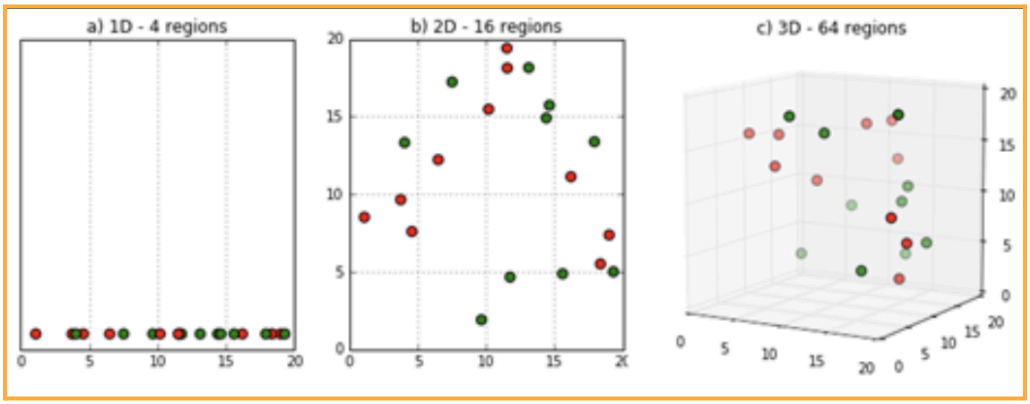
- **Y** is the dependent variable (the outcome you want to predict).
- X_1, X_2, X_3, are the independent variables that you believe influence **Y**.
- β_0 is the intercept, representing the expected value of **Y** when all **Xs** are 0.
- $\beta_1, \beta_2, \beta_3$, are the coefficients for each independent variable, quantifying their individual impact on **Y**.
- ϵ represents the error term, accounting for the variation in **Y** not explained by the **X**s.

Higher Dimensionality Discussions in Multiple Linear Regression

As the number of predictors in a multiple linear regression model increases, certain issues become more pronounced, affecting the overall performance and interpretability of the model. Here we discuss three key challenges related to higher dimensionality: the Curse of Dimensionality, the inadequacy of R^2, and multicollinearity.

1. Curse of Dimensionality

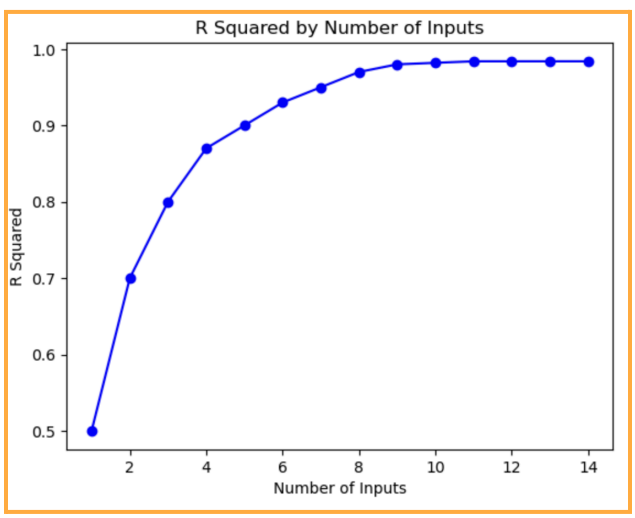
Overview: The Curse of Dimensionality refers to various phenomena that arise when analyzing and organizing data in high-dimensional spaces that do not occur in low-dimensional settings.

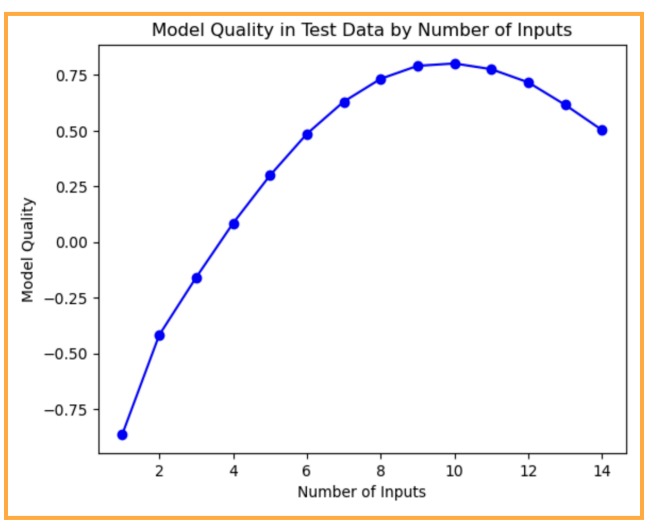


2. R^2 Not Adequate for Multiple Linear Regression

Overview: In the context of multiple linear regression with many predictors, **R^2** alone may not always provide a complete picture of the model's performance:

- **Inflation of R^2**: R^2 tends to increase as more variables are added to the model, regardless of whether those variables are relevant to the outcome. This can mislead the interpretation of model improvement.
- **Adjusted R^2**: Unlike R^2, Adjusted R^2 accounts for the number of predictors in the model and adjusts the statistic based on the complexity, providing a more accurate measure of model performance.





Adjusted R^2

Adjusted R^2 is R^2 penalized by the number of inputs (k):

$$AdjR^2 = 1 - \frac{SSE/(n-k)}{SST/(n-1)}$$

Multicollinearity

Overview: Multicollinearity occurs when two or more predictors in a multiple regression model are highly correlated. This can lead to several problems:

Coefficient Instability: Highly correlated predictors can make the coefficient estimates very sensitive to small changes in the model, leading to high variance in the coefficient estimates.

Inflated Standard Errors: This can inflate the standard errors of the coefficients, leading to a failure in detecting significant predictors.

Consequences of Multicollinearity

With multicollinearity, the following problems arise:

- Impossible or difficult to calculate an inverse of $\mathbf{X}^T\mathbf{X}$
- Difficult to interpret coefficient estimates
- Inflated standard errors on coefficient estimates

Dimension Reduction

Dimension reduction is a critical process in data preprocessing, especially in fields dealing with high-dimensional data such as machine learning, statistics, and pattern recognition. Reducing the dimensionality of a dataset can simplify models, enhance performance, and reduce computational costs. Let's explore why dimension reduction is necessary, some common feature selection methods, and linear algebra techniques like Principal Component Analysis **(PCA)**.

Need for Dimension Reduction

Feature Selection

Feature selection involves choosing a subset of relevant features for model construction without transforming them. It improves model interpretability, reduces overfitting, and enhances generalization by removing irrelevant or redundant predictors.

Linear Algebra Methods: PCA

Principal Component Analysis (PCA) is a technique that transforms the original variables into a new set of variables, which are linear combinations of the original variables. These new variables (principal components) are orthogonal and ranked according to the variance of data along them.

- **How PCA Works**:
  - **Standardization**: Standardize the range of the continuous initial variables so that each one of them contributes equally.
  - **Covariance Matrix Computation**: Compute the covariance matrix to understand how the variables vary from the mean with respect to each other.

- **Compute Eigenvalues and Eigenvectors**: The eigenvectors of the covariance matrix are the directions of the axes where there is the most variance (most information). Eigenvalues represent the amount of variance carried in each Principal Component.
- **Feature Vector Formation**: Eigenvectors are sorted by eigenvalues in descending order so that the first few retain most of the variation present in all of the initial variables.

Feature Selection Methods

Sequential methods:

Backward Elimination (Backward Selection): - Start with all candidate features. Iteratively remove the least significant feature (the one with the highest p-value in regression analysis), one at a time, until a specified stopping criterion is reached (like a set number of features, or a threshold for model performance).

Forward Selection: - Start with no variables in the model. In each step, add the variable that improves the model the most until adding a new variable does not improve the model by a significant amount.

Correlation Ranking - Covariance & Correlation

Correlation ranking is a vital technique used in the process of feature selection and data analysis to identify relationships between variables. This involves understanding both covariance and correlation, which are statistical measures that describe how two variables are related.

Covariance

Definition:

- Covariance measures the joint variability of two random variables. It indicates the direction of the linear relationship between variables.

Calculation:

- The covariance between two variables X and Y is calculated as: $\mathrm{Cov}(X, Y) = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})}{n-1}$ Where:
- x_i and y_i are individual sample points indexed with i.
- $\overline{x}$ and $\overline{y}$ are the sample means of X and Y, respectively.
- n is the number of data points.

Interpretation:

A positive covariance indicates that as one variable increases, the other variable tends to increase.

A negative covariance indicates that as one variable increases, the other tends to decrease.

Zero covariance indicates that the variables are independent (Note: This only implies lack of linear dependence).

Correlation

Definition:

Correlation, specifically the Pearson correlation coefficient, standardizes the covariance by the standard deviations of the variables, giving a dimensionless value that measures the strength and direction of the linear relationship between two variables.

Calculation:

- The correlation coefficient between two variables X and Y is calculated as: $r = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$
  Where:
- σ_X and σ_Y are the standard deviations of X and Y, respectively.

Interpretation:

- **Value range**: $-1 \leq r \leq 1$
  - $r = 1$: Perfect positive linear relationship.
  - $r = -1$: Perfect negative linear relationship.
  - $r = 0$: No linear relationship.
- The magnitude of r indicates the strength of the relationship. The closer r is to either -1 or 1, the stronger the linear relationship.

Using Covariance and Correlation for Feature Selection

Correlation Ranking:

By calculating the correlation coefficients between the dependent variable and each independent variable, you can rank the features based on the absolute values of their correlation coefficients.

Features with higher correlation to the outcome variable are often considered more important and are selected preferentially.

Pearson and Spearman Correlation

Correlation is a statistical measure that expresses the extent to which two variables are linearly related. It's a crucial tool in data analysis for understanding relationships between variables, though it's essential to remember that correlation does not imply causation. Here's a closer look at the two common types of correlation: Pearson and Spearman, along with visualizing correlation signs.

Definition:

Pearson correlation (also known as the Pearson product-moment correlation coefficient) quantifies the degree to which a linear relationship exists between two variables.

Calculation:

- The Pearson correlation coefficient, r, is calculated as: $r = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum(x_i - \overline{x})^2 \sum(y_i - \overline{y})^2}}$
- Here, x_i and y_i are the individual sample points, and $\overline{x}$ and $\overline{y}$ are the means of the x and y samples, respectively.

Spearman Correlation

Definition:

- Spearman correlation assesses how well the relationship between two variables can be described using a monotonic function. It's based on the ranks of the values rather than the raw data.

Calculation:

- The Spearman correlation coefficient, ρ, is determined similarly to Pearson but using the rank values of x and y: $\rho = \frac{\sum(rank(x_i)-\overline{rank(x)})(rank(y_i)-\overline{rank(y)})}{\sqrt{\sum(rank(x_i)-\overline{rank(x)})^2 \sum(rank(y_i)-\overline{rank(y)})^2}}$

Principal Component Analysis (PCA) - Intuition and Explanation

Principal Component Analysis (PCA) is a powerful statistical technique used for dimensionality reduction while attempting to maintain as much information as possible. It's particularly useful in cases where you have numerous interrelated variables, and you want to simplify them into a smaller set that still captures the crucial information.

Why Use PCA?

Imagine you're analyzing a dataset about cars, and you have several variables like color, size, weight, number of cylinders, horsepower, and MPG. Many of these attributes are correlated; for example, weight and size often increase together, and horsepower could correlate with the engine size (number of cylinders). PCA helps to reduce the dimensionality of such data by finding new variables that are linear combinations of the original variables but are uncorrelated (orthogonal) to each other.

How Does PCA Work?

PCA transforms the original correlated variables into a new set of variables, which are linear combinations of the original variables. These new variables, called principal components (PCs), are orthogonal, ensuring they are linearly independent.

1. **Standardization**: The first step in PCA is to standardize the data. This involves scaling each variable to have a mean of zero and a standard deviation of one. This is crucial because it puts all variables on the same scale, preventing variables with larger scales from dominating the analysis.
2. **Covariance Matrix Computation**: PCA computes the covariance matrix of the data to understand how the

variables vary from the mean with respect to each other. The covariance matrix provides a measure of the strength and direction of the correlation between pairs of variables.

3. **Eigenvalue Decomposition**: The next step is to decompose the covariance matrix into its eigenvectors and eigenvalues. This decomposition helps to understand the directions of the maximum variance (eigenvectors) in the data, and the magnitude of these directions (eigenvalues).
4. **Selecting Principal Components**: The eigenvectors that correspond to the largest eigenvalues are selected as the principal components because they capture the most variance in the data. The number of principal components selected depends on the amount of variance you want to retain in your reduced dataset.
5. **Transformation**: The original data are transformed into a new coordinate system defined by the selected eigenvectors. The transformed data are the scores of the original data on the principal components.

Covariance Matrix and Eigen Decomposition in PCA

Understanding the Covariance Matrix and Eigen Decomposition is fundamental to performing Principal Component Analysis (PCA). These mathematical concepts allow us to identify new directions in the feature space that best summarize the data. Here's a deeper dive into each component and its role in PCA.

Covariance Matrix

The covariance matrix is a key mathematical concept in statistics and data analysis that captures the covariance (i.e., the measure of how much two random variables change together) between pairs of elements in a data set.

- **Purpose**: It helps in understanding how each variable in a dataset varies from the mean with respect to every other variable. This is crucial in PCA because it identifies the directions with the maximum variance.
- **Calculation**: For a data matrix **X** of size n×p (where **n** is the number of observations and **p** is the number of variables), the covariance matrix **C** is a p×p matrix where each element represents the covariance between two variables.

Eigen Decomposition of the Covariance Matrix

Eigen Decomposition is a type of matrix decomposition in which a matrix is broken down into its constituent elements in such a way that it reveals the underlying properties of the matrix. In the context of PCA, it involves decomposing the covariance matrix into eigenvectors and eigenvalues.

- **Eigenvectors**: These are the directions or principal components in the feature space. Eigenvectors of the covariance matrix represent the axes of the new feature space that explain the variance of the data.

- **Eigenvalues**: Each eigenvector has a corresponding eigenvalue, which quantifies the amount of variance (or "spread") captured by its eigenvector. A higher eigenvalue indicates that the eigenvector captures more variance.

PCA Steps Involving Covariance and Eigen Decomposition

1. **Standardize the Data**: Scale the data so each feature contributes equally.
2. **Compute the Covariance Matrix**: Reflects how changes in one variable are associated with changes in another.
3. **Eigen Decomposition**: Decompose the covariance matrix to find its eigenvectors and eigenvalues.
4. **Sort Eigenvectors**: The eigenvectors are sorted by descending eigenvalues. The eigenvector with the highest eigenvalue is the direction along which the data is most spread out.
5. **Project Data**: Project the original data onto the new feature space using the selected eigenvectors (those corresponding to the largest eigenvalues).

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Applications and Further Insights

- **Eigenface for Image Recognition**: PCA can be used in image recognition tasks where images, treated as high-dimensional data points, are transformed into a lower-dimensional space using PCA (termed as Eigenfaces in facial recognition). This simplifies the dataset while retaining essential features needed for recognition.
- **Data Recovery**: PCA allows for the reconstruction of the original variables from the reduced feature space by an inverse transformation, which can be useful for data compression and recovery.

Rotation of Coordinates

- **Relook**: PCA can be viewed as a rotation of the coordinate system used to describe the data. The original axes (variables) are rotated to align with the directions of maximum variance, defined by the eigenvectors of the covariance matrix.

Additional Resources

- **Books:**
  - Introduction to Linear Algebra by Gilbert Strang.
  - The Elements of Statistical Learning by Hastie, Tibshirani, and Friedman.
- An intuitive explanation of vectors | by Skyler Dale | Towards Data Science
- 3) Matrices - Linear Algebra
- Properties of Matrices Transpose with Examples and Solutions
- Inverses - Linear Algebra
- Eigen Intuitions: Understanding Eigenvectors and Eigenvalues
- Real life example of Eigen values and Eigen vectors

Conclusion

Linear Algebra and Regression are critical tools in machine learning and data science, enabling us to model relationships, reduce dimensions, and perform essential mathematical operations for analyzing data. By mastering these topics, you will build a strong foundation for understanding and applying more advanced techniques in the field.